L-TORSION IN THE PICARD GROUP OF A PROPER CURVE IS BOUNDED BY THE FIRST COHOMOLOGY PLUS THE GEOMETRIC GENUS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the base-change reduction, the H^1 surjection onto the reduction with its vanishing justification, the component and torsion isomorphisms, and the final proposition citation are all retained verbatim.

Editorial note. The label lemma-bound-torsion-simple names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Picard and cohomology under base change. Varieties, Lemma varieties-lemma-change-fields-pic; Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology. Injects Pic(X) into Pic(X_{overline{k}}) and preserves dim H^1 under the field base change, reducing to the algebraically closed case.
- Reduction to the reduced curve. Cohomology, Proposition cohomology-proposition-vanishing-Noetherian; Picard Schemes of Curves, Lemma pic-lemma-torsion-descends. The reduction surjection stays surjective on H^1 by Cech vanishing, and X_red to X is an isomorphism on components and on l-torsion of Picard groups.
- The reduced-proper torsion proposition. Same chapter, Proposition proposition-torsion-picard-reduced-proper. Supplies the final bound dim Pic(X)[ell] <= dim H^1(O_X) + g_geom for reduced proper curves over algebraically closed fields.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Prerequisite pointer invoked at proof line(s) 3972; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 02UZ: sources/stacks-chow/tags.txt, source lines 3750–3750. Prerequisite pointer invoked at proof line(s) 3983; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CD5: sources/stacks-chow/tags.txt, source lines 15530–15530. Prerequisite pointer invoked at proof line(s) 3987; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C20: sources/stacks-curves/pinned-original-source.tex, source lines 3701–3701. Prerequisite pointer invoked at proof line(s) 3989; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CC5: sources/stacks-chow/tags.txt, source lines 15495–15495. Prerequisite pointer invoked at proof line(s) 3971; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme of dimension ≤ 1 over k. Let ℓ be a prime number invertible in k. Then*

$$\dim_{\mathbf{F}_\ell} \mathrm{Pic}(X)[\ell] \leq \dim_k H^1(X, \mathcal{O}_X) + g_{geom}(X/k)$$

where $g_{geom}(X/k)$ is as defined above.

Proof. The map $\mathrm{Pic}(X) \to \mathrm{Pic}(X_{\overline{k}})$ is injective by Varieties, Lemma 0CC5. By Cohomology of Schemes, Lemma 02KH $\dim_k H^1(X, \mathcal{O}_X)$ equals $\dim_{\overline{k}} H^1(X_{\overline{k}}, \mathcal{O}_{X_{\overline{k}}})$. Hence we may assume k is algebraically closed.

Let X_{red} be the reduction of X. Then the surjection $\mathcal{O}_X \to \mathcal{O}_{X_{red}}$ induces a surjection $H^1(X, \mathcal{O}_X) \to H^1(X, \mathcal{O}_{X_{red}})$ because cohomology of quasi-coherent sheaves vanishes in degrees ≥ 2 by Cohomology, Proposition 02UZ. Since $X_{red} \to X$ induces an isomorphism on irreducible components over $\overline{k}$ and an isomorphism on ℓ-torsion in Picard groups (Picard Schemes of Curves, Lemma 0CD5) we may replace X by X_{red}. In this way we reduce to Proposition 0C20. $\square$